

Bradley Nguyen

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EDUCATION

Carleton University, Bachelor of Computer Science (Co-op)

Ottawa, ON

Major CGPA: 10.19/12.0

2023 – 2027

Relevant Coursework: Abstract Data Types and Algorithms, Object-Oriented Software Engineering, Web Development, Programming Paradigms, Database Management Systems, Operating Systems

EXPERIENCE

Innovation, Science and Economic Development Canada

May 2026 – Present

Data Engineer | Co-op | AWS, Python, Oracle SQL

Ottawa, ON

- Developed ETL scripts to load, transform, and validate **3+ source datasets** into staging tables, reducing manual processing by **40%** and improving daily data delivery reliability
- Automated SQL and shell-based batch jobs for scheduled data refreshes, improving runtime efficiency by **25%**
- Implemented end-to-end ETL validation and logging on **EC2-hosted runners and AWS Lambda** to catch load errors and improve release reliability

Lux Cognitiva Technologies Inc.

May 2026 – Present

Software Engineer | Contract Part-Time | TypeScript, Prisma, PostgreSQL, AWS

Remote

- Built a **Health Canada DPD** sync and search API using **TypeScript, Prisma, and PostgreSQL**, with migrations, Zod validation, and integration tests
- Developed scheduling APIs with **6 Prisma models, 6 routes**, RBAC/audit enforcement, and backend test coverage for visit workflow logic

dynaCERT Inc.

May 2025. – Dec. 2025

Full-Stack Developer | Co-op | TypeScript, React, Node.js, Express.js, C++, MySQL, Docker

Toronto, ON

- Architected and deployed a **React/TypeScript** industrial IoT dashboard (**Node.js + MySQL**), processing real-time telemetry for **450+ H₂** generator units
- Built a **real-time telemetry ETL pipeline** for H₂ generators, parsing **40+ sensor fields** from **ZeroMQ streams** into MySQL with validation and error handling.

PROJECTS

MediaspaceDownloader | [GitHub](#) | Python, Playwright, Selenium, FFmpeg

Jan. 2026 – Apr. 2026

- Developed a **Python video downloader** that extracts **Kaltura M3U8 links** using **7 URL patterns, 7 HTML regex patterns**, and **6 API endpoint variations** to handle multiple Mediaspace URL formats
- Implemented **browser automation** with **2 frameworks** (Selenium and Playwright) to bypass download restrictions, achieving **2 minute** capture time

Dumpster Diver | [GitHub](#) | Python, TensorFlow, Keras, Arduino, SQLite, Flask, PHP

Mar. 2025

- **Winner** at **cuHacking 6** (\$300, 300+ participants) for an **IoT-enabled waste sorting system** using **TensorFlow CNNs, Flask APIs**, and **Arduino servos** for automated material classification, processing **100+ items/min** with **95% accuracy**
- Trained **TensorFlow CNN** for object detection, connected to Arduino servos via Flask API

EXTRACURRICULAR ACTIVITIES

cuHacking

Sep. 2025 – Jun. 2026

Hacker Experience Team Lead

Ottawa, ON

- Coordinate and enhance the participant experience for **300+ hackers** through event planning, workshops, and sponsor engagement initiatives

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL (Postgres), JavaScript, TypeScript, HTML/CSS

Frameworks/Libraries: React, Next.js, Node.js, Express.js, Three.js, Flask, Tailwind, Vite, MongoDB, Selenium

Developer Tools: Git, GitHub, Docker, GCP, Linux/Unix, Jira, Vercel, Power BI, Azure, AWS, Playwright, OAuth